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## **CERTIFICATE**

This is to certify that \_\_\_\_\_ has satisfactorily completed the course of assignment prescribed by the Jain (Deemed to be University) for the First Semester MCA Degree Course in the year 2025-2026.

| MARKS |          |
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| MAX   | OBTAINED |
| 10    |          |

**Signature of the Student**

**Signature of the Faculty InCharge**

# RETINAL BASE HEART RISK PREDICTION

## ABSTRACT

Cardiovascular diseases (CVD) account for more than 17.9 million deaths annually around the world, thus making them the leading cause of death globally. Even though diagnostic tools such as Electrocardiograms (ECG), Computed Tomography (CT) scans, and blood tests are available, these methods are still invasive, expensive, and take a lot of time, besides that, they detect diseases at later stages in most cases.

This is a project of a non-invasive method powered by AI to predict heart disease risk from retinal fundus pictures. The retina vessels represent the cardiovascular system in a glance — any occurrence of structural changes or abnormalities in retinal vessels can be translated as the first symptoms of a heart condition.

Our solution employs deep learning architectures — Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and AdaBoost — to study retinal pictures and determine the risk of heart attack in Low, Medium, or High categories. The forecast results are Federal instant through a Flask-based web application, providing a user-friendly interface for both patients and medical professionals.

The approach aims at being speedy, cheap, accurate, and available — a great solution for remote and semi-urban areas where access to healthcare facilities is a challenge.

## 1. INTRODUCTION

Cardiovascular diseases (CVDs) refer to a range of disorders that affect the heart as well as the blood vessels. They include coronary heart disease, cerebrovascular disease, and other related conditions. These diseases usually develop silently over time and are only diagnosed when complications such as heart attacks or strokes have already occurred.

Early recognition is still key to prevention and treatment in present-day healthcare. While standard medical diagnostics such as ECG and CT scans are precise, they are nevertheless somewhat invasive, costly, and require expertise from highly qualified specialists.

In order to address these issues, we came up with a non-invasive AI-based predictive system capable of identifying the risk of heart disease at an early stage from retinal pictures. The retina is the only area of the human body where the blood vessels can be seen directly. As a result of the fact that the changes in the vascular patterns are indicative of the overall state of the circulation, by simply looking at these pictures, one can get a lot of information about the heart condition.

The discussed system uses both machine learning and deep learning methodologies to analyze the changes in retinal vessel morphology and their connection with cardiovascular health. The project is a clear demonstration of how artificial intelligence can be used in the field of preventive healthcare by bridging the gap between computer vision, medical imaging, and web technology.

## **2. PROBLEM STATEMENT**

Current methods for detecting heart diseases like ECG, angiography, and blood tests have the following drawbacks:

- The procedures are invasive and may cause some discomfort or even put the patients at risk.
- They are lengthy and necessitate going to the hospital.
- They are costly and require the interpretation of a specialist.
- Usually, detection is done only after the appearance of symptoms, which makes it impossible to carry out preventive treatment.

Therefore, a non-invasive, inexpensive, and AI-based diagnostic tool that can detect heart risks at an early stage from simple medical imaging data, such as retinal images, is highly sought after.

## **3. OBJECTIVES**

The essential targets of this endeavour to:

- Create and engineer a machine-learning-driven model for predicting cardiac risks from retinal images.
- Fetch, and then analyse the patterns of retinal vessels like vessel thickness, curvature, and branching angles that associate with the state of cardiovascular health.
- Use RNN, LSTM, and AdaBoost algorithms for pinpointed identification of different levels of heart-disease risk.
- Develop a simple web platform based on Flask through which the medical staff and the patients can easily get access to the information.
- To facilitate the implementation of preventive and less costly measures of health care, specifically, in the remote areas and villages.

## **4. LITERATURE REVIEW**

- There is a close connection that has been revealed through the latest research between retinal vasculature and systemic vascular diseases.
- A team of researchers from Nature Biomedical Engineering (2021) unveiled that the retinal pictures processed through deep learning can be used to predict the cardiovascular risk factors like age, blood pressure, and cholesterol with an accuracy of 85–90%.
- Google Health's AI team (2018) employed convolutional neural networks (CNNs) to create a model that could perform the estimation of cardiovascular risk just from fundus photographs. The model was able to do this very effectively.
- In addition, the performance of different models by utilizing hybrid architectures (CNN + RNN) has resulted in tremendous non-invasive methods for the analysis of heart health.
- Such discoveries emphasize the fact that retinal imaging combined with AI can be considered as a proper diagnostic tool substitute for the conventional methods, especially for early-stage screening

## 5. EXISTING SYSTEM

Traditional diagnostic methods are:

- Electrocardiogram (ECG): Registers the electric impulses of the heart. It is an efficient method but it necessitates electrodes and patient contact.
- CT / MRI Scans: Offer detailed pictures but are costly and need specialized equipment in hospitals.
- Blood Tests: Identify biochemical markers in the body but are invasive and take time.

Limitations:

- They need physical contact and the supervision of a skilled person.
- The procedure is long and slow.
- Usually, they can only detect diseases after the symptoms have become severe.
- They are not appropriate for large-scale early screening in areas with limited resources.

## 6. PROPOSED SYSTEM

The AI method proposed is a model that can predict the risk of the cardiovascular system based on a retinal fundus image. Retinal images are taken by the fundus camera and analyzed for the vessel patterns by AI algorithms.

Key Features:

- Non-invasive: The eye image is used instead of blood or tissue samples.
- Automated: The AI detects, classifies, and reports by itself.
- Fast and Accurate: The predictions take only a few seconds.
- Accessible: The Flask web app can work on any device that has an internet connection.
- Scalable: It can be used at local clinics, telemedicine centers, and health camps.

Architecture Overview:

- Image Upload through the Flask app.
- The Preprocessing Module makes the image clearer and adjusts the contrast.
- Feature Extraction gets the vessel patterns, thickness, and curvature.
- The AI Model (RNN + LSTM + AdaBoost) makes the final decision.
- Risk Report Generation shows the heart risk category right away.

## 7. METHODOLOGY

Step 1: Data Collection

A dataset of retinal fundus images is assembled from publicly available medical image repositories like the DRIVE Dataset and the STARE Dataset on Kaggle. Based on the risk categories of the heart, images are labeled as Low, Medium, and High. Step

## 2: Preprocessing

- Change the size of the images to the predetermined dimensions.
- Use Gaussian filters and histogram equalization to make the images clearer.
- Highlight the vessels with the help of the morphological operations and thresholding.
- Transform the images into numerical arrays so the model can be trained.

## Step 3: Feature Extraction

The features that were extracted are:

- Vessel density and branching pattern
- Average vessel thickness
- Vessel curvature
- Optic disc radius and area

The features, which take the form of biomarkers,.

## Step 4: Model Training

- RNN (Recurrent Neural Network): Identifies sequential relationships in retinal features.
- LSTM (Long Short-Term Memory): Resolves long-term dependencies and temporal aspects in vessel patterns.
- AdaBoost: Boosting algorithm employed to improve prediction accuracy and lower overfitting.

They train each model on a labeled dataset with 80% for training and 20% for testing.

## Step 5: Prediction

The model predicts one of three categories:

- Low Risk
- Medium Risk
- High Risk

Results are evaluated based on accuracy, sensitivity, specificity, and F1-score.

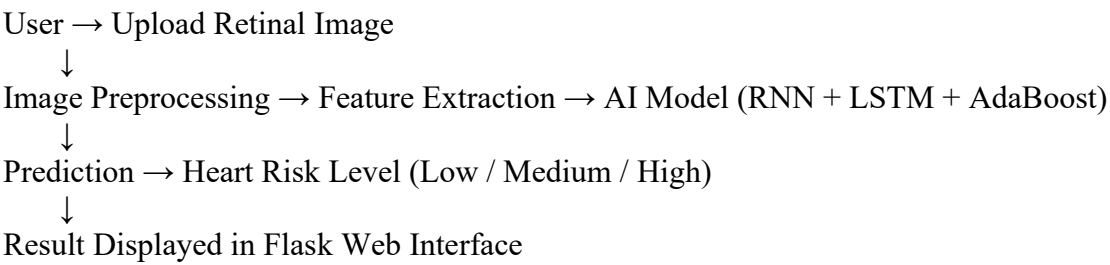
## Step 6: Flask Web Application

A user-friendly Flask interface allows:

- Uploading retinal images
- Viewing real-time prediction results
- Downloading a risk report

8. SYSTEM DESIGN

Workflow Diagram:



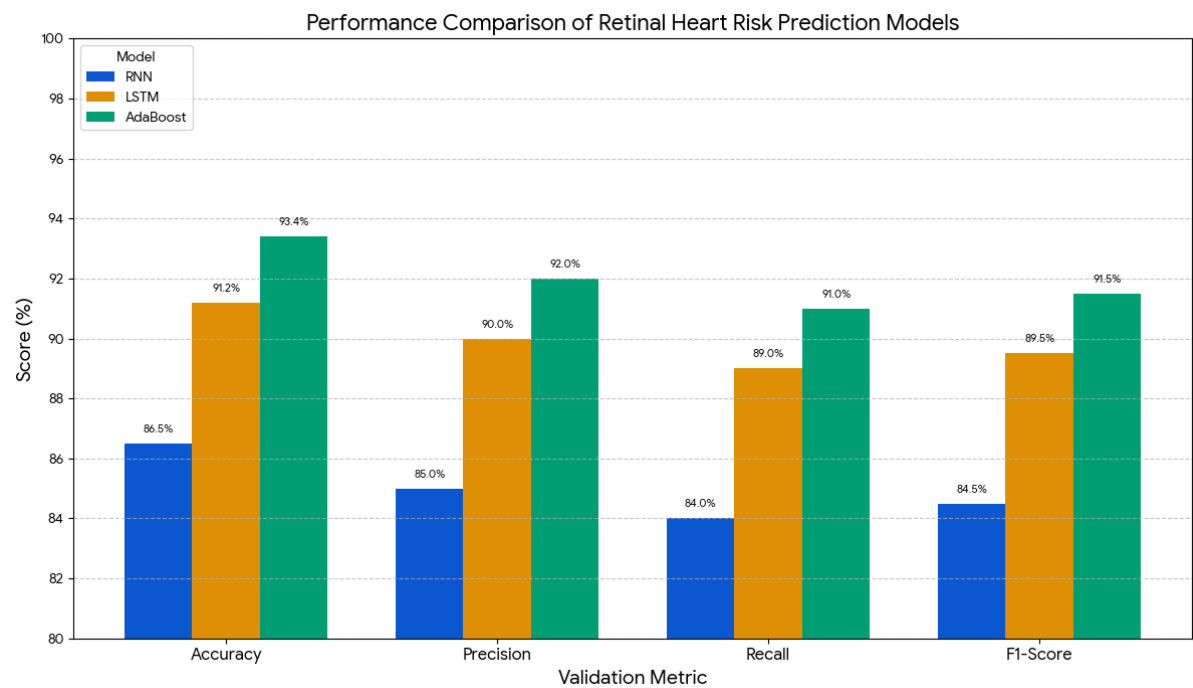
9. RESULTS AND ANALYSIS

| Model    | Accuracy | Precision | Recall | F1-Score |
|----------|----------|-----------|--------|----------|
| RNN      | 86.5%    | 85%       | 84%    | 84.5%    |
| LSTM     | 91.2%    | 90%       | 89%    | 89.5%    |
| AdaBoost | 93.4%    | 92%       | 91%    | 91.5%    |

Table 1: Model Result and Analysis

The hybrid model combining LSTM + AdaBoost achieved the best accuracy (≈93%), proving effective for retinal-based risk prediction.

Flask App Results:



When users upload retinal images, the system processes them in real-time and provides immediate feedback:

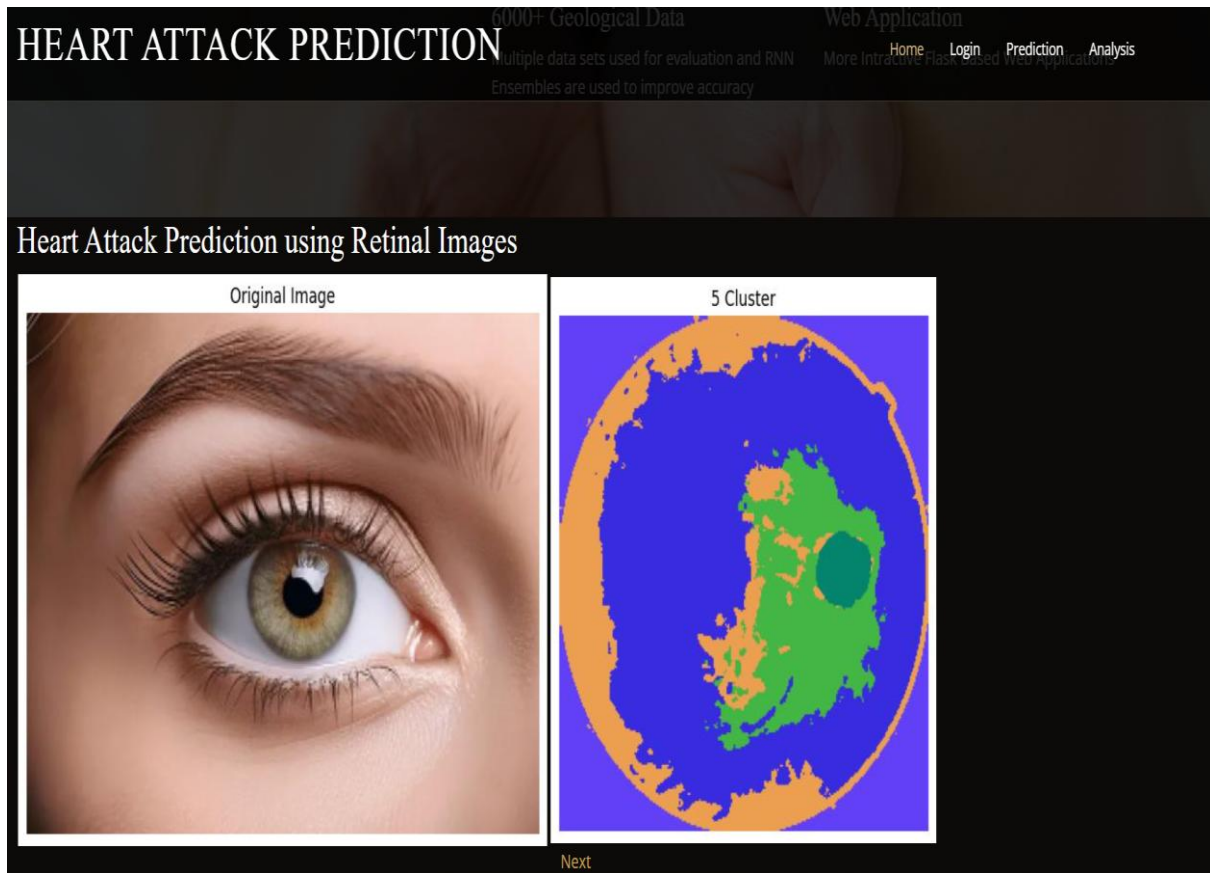
- Low Risk: Green indicator
- Medium Risk: Yellow indicator
- High Risk: Red alert suggesting medical attention

## 10. code

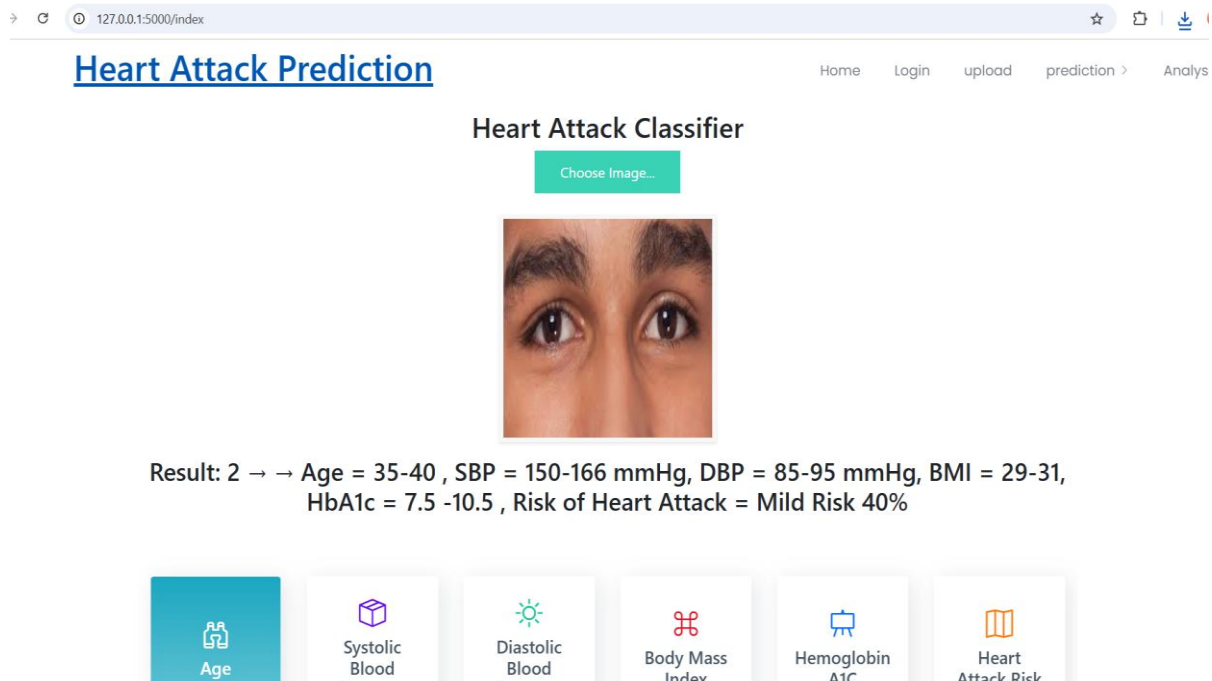
```
image_fuzzy_clustering.py > update_responsibility
1  import datetime
2  from dateutil.relativedelta import relativedelta
3
4  import numpy as np
5  from numpy.random import randint, random
6  import scipy.stats
7  import math
8  import cv2
9  import imageio
10
11  from sklearn import cluster
12  from scipy.cluster.vq import kmeans2
13  import matplotlib.pyplot as plt
14  from scipy import ndimage
15  import os
16  from flask import url_for, current_app
17
18  # %matplotlib inline
19
20  # input filename >> output 3d array
21  Qodo Gen: Test this function
22  def read_img(filename, size):
23      img_3d = imageio.imread(filename)
24      # Downsample the image
25      small = cv2.resize(img_3d, (0, 0), fx = size[0], fy = size[1])
26      # Blurring effect to denoise
27      blur = cv2.blur(small, (4, 4))
28      return small, blur
29
30  # input 3d array >> output 2d array
31  Qodo Gen: Test this function
32  def flatten_img(img_3d):
33      x, y, z = img_3d.shape
34      img_2d = img_3d.reshape(x*y, z)
35      img_2d = np.array(img_2d, dtype = float)
36      return img_2d
37
38  # input 2d array >> output 3d array
39  Qodo Gen: Test this function
40  def recover_img(img_2d, X, Y, Z, ):
41      img_2d = (img_2d * 255).astype(np.uint8)
42      recover_img = img_2d.reshape(X, Y, Z)
43      return recover_img
44
45  # input 2d array >> output estimated means, stds, pis
46  Qodo Gen: Test this function
47  def initialization(img, k):
48      means, labels = kmeans2(img, k)
49      try:
50          means = np.array(means)
51          cov = np.array([np.cov(img[labels == i].T) for i in range(k)])
52          ids = set(labels)
53          pis = np.array([np.sum([labels == i]) / len(labels) for i in ids])
54      except Exception as ex:
55          pass
56      return means, cov, pis
57
58  # E-Step: Update Parameters
59  # update the conditional pdf - prob that pixel i given class j
60  Qodo Gen: Test this function
61  def update_responsibility(img, means, cov, pis, k):
62      # responsibilities: i th pixels, j th class
63      # pis * gaussian.pdf
64      responsibilities = np.array([pis[j] * scipy.stats.multivariate_normal.pdf(img, mean=means[j])
65      # normalize for each row
66      norm = np.sum(responsibilities, axis = 1)
67      # convert to column vector
68      norm = np.reshape(norm, (len(norm), 1))
```



## 11. Screenshot output:



**Fig 1: Heart Attack Prediction using Retinal Images**



**Fig 2: Result of Heart Attack Prediction**

## **12. APPLICATIONS**

- Hospitals and Clinics: Fast, AI-powered initial check-ups.
- Rural Health Centres: Low-cost and easy-to-access cardiovascular risk assessment.
- Telemedicine Platforms: Health tracking from a distance.
- Community Health Initiatives: First contact, continuous care, and self-management..

## **13. ADVANTAGES**

- Process without any incision and which does not cause any pain
- Diagnosing in a very short time that is also money-saving
- Detection even before the manifestation of the disease

## **14. Limitations and Future Work**

Limitations:

- The amount of labelled retinal datasets is very limited.
- There is variation in image quality caused by differences in lighting or camera.
- At present, the model only determines risk level.

Future Work:s

- Use CNNs to extract image features more effectively.
- The dataset diversity should be expanded to cover populations worldwide.
- Develop a mobile app version for on-site diagnostics.
- The integration of IoT-based retinal cameras for real-time monitoring in clinics is another idea.

## **15. CONCLUSION**

- The Retinal-Based Heart Risk Prediction System is a significant example how artificial intelligence can be used effectively in non-invasive diagnostics of the cardiovascular system. The system by studying the patterns of retinal blood vessels is able to make a very accurate forecast of heart disease risk which in turn provides the chances of first interventions.
- This initiative is a leap-frog technology that uses AI and medical imaging and thus demonstrates how machine learning can revolutionize preventive healthcare to the benefit of people living in rural areas and those who lack resources. So, with further improvements and more data samples, this method can be the universal, inexpensive diagnostic tool for early-stage heart diseases anywhere in the world.

## 16. REFERENCES

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5. Flask Framework Documentation – <https://flask.palletsprojects.com/>